A recursive digital filter uses previous output samples along with the current to calculate the next output. A general recursive filter can be written as.

$$y(n) = \sum b_k x(n-k) + \sum a_k y(n-k)$$

Since Q15 is a fixed point representation the output has a limited numerical range

Q15 uses 16 bits:

1 bit → sign

15 bit → fraction

Range $-1 \le x < 1$

The max postive value is:

0.9999695

The min value is

-1

Therefore an output is larger than +1 or smaller than -1 cannot be directly represented in Q15.

Observed Error

If the recursive filter produces

$$|y(n)| > 1$$